



Command Cheatsheet - for practice

Networks and switching (Swinburne University of Technology (Vietnam))



Scan to open on Studeersnel

Command Cheatsheet

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1. Display Interface Status

```
Switch# show ip interface brief
```

2. Test Network

```
>ipconfig
```

```
>ping 192.168.1.11
```

3. Address Resolution Protocol (ARP)

```
>arp -a
```

```
>arp -d
```

- Ping again to validate the ARP tables

4. CLI Operation Modes

- Administrator

```
>enable  
#exit  
>
```

- Configuration mode

```
>enable  
#config t  
(config)#end  
#exit  
>
```

5. Device Name(hostname)

```
Switch>ena
Switch#conf t
Switch(config)#hostname Switch3
Switch3(config)#end
Switch3#
```

6. MOTD

```
Switch3#conf t
Switch3(config)#banner motd + *****
** This is Week 2 **
** of Semester 2 **
** at Swinburne **
***** +
Switch3(config)#end
Switch3#
```

7. Saving configuration

```
#show running-config
#show startup-config
#copy running-config startup-config
#reload
```

8. Password

- Set password to User Mode

```
Switch3#conf t
Switch3(config)#line console 0
Switch3(config-line)#password ccna
Switch3(config-line)#login
```

```
Switch3(config-line)#end
Switch3#exit
```

- Delete password

```
Switch3#conf t
Switch3(config)#line console 0
Switch3(config-line)#no password
Switch3(config-line)#no login
Switch3(config-line)#end
Switch3#exit
```

- Set password to Admin Mode

```
Switch3#conf t
Switch3(config)#enable enable secret
```

- Delete password

```
Switch3#conf t
Switch3(config)#no enable secret
```

9. Configure Ethernet Interfaces

- Disable Ethernet Interfaces

```
Switch4#conf t
Switch4(config)# interface gigabitEthernet 1/0/1
Switch4(config-if)#shutdown
Switch4(config)# interface range gigabitEthernet 1/0/2 - 4
Switch4(config-if)#shutdown
Switch4(config-if)#end
```

- Enable again

```
Switch4#conf t
Switch4(config)# interface gigabitEthernet 1/0/1
```

```
Switch4(config-if)#no shutdown
Switch4(config-if)#end
```

10. Configure the Switch Management Interface

```
Switch3(config)interface vlan1
Switch3(config-if)#ip address 192.168.1.3 255.255.255.0
Switch3(config-if)#no shutdown
Switch3(config-if)#end
```

- Use the ping command on switches to validate that

```
#ping 192.168.1.4
```

11. Cleaning up

- Delete startup-configuration

```
Switch3#write erase
```

- Delete vlan database

```
show flash
```

```
Switch#delete vlan.dat
```

12. Observe VLAN Configuration

```
#show vlan brief
```

13. Configure User Vlans

```
Switch3#config t
Switch3(config)#vlan 10
Switch3(config-vlan)#name Users
Switch3(config-vlan)#end
```

14. Configure VLAN Membership

```
Switch3#config t
Switch3(config)#interface range gigabitEthernet 1/0/10 - 14
Switch3(config-if-range)#description Users VLAN
Switch3(config-if-range)#switchport mode access
Switch3(config-if-range)#switchport access vlan 10
```

15. Configure Switch Management

- Disable Interface VLAN1

```
Switch3#config t
Switch3(config)#interface Vlan 1
Switch3(config-if)#shutdown
Switch3(config-if)#exit
```

- Create management VLAN99

```
Switch3(config)#vlan 99
Switch3(config-vlan)#name Management
Switch3(config-vlan)#exit
```

- Create management Interface VLAN99 and allocate the management IP

```
Switch3(config)#interface vlan 99
Switch3(config-if)#description Management
Switch3(config-if)#ip address 192.168.99.3 255.255.255.0
Switch3(config-if)#exit
```

16. Telnet Access

```
Switch3(config)#line vty 0 15
Switch3(config-line)#password ccna
Switch3(config-line)#login
```

- Configure VLAN trunking

```
Switch3(config)#interface range gigabitEthernet 1/0/5 - 6
Switch3(config-if-range)#switchport mode trunk
```

- Telnet access to other switch

```
Switch4#telnet 192.168.99.3
```

- EXIT to back to CLI
- Disable Telnet Access

```
Switch3(config)#line vty 0 15
Switch3(config-line)#transport input none
Switch3(config-line)#no login
Switch3(config-line)#end
```

17. SSH Access

- Configure a domain name

```
Switch3(config)#ip domain-name ccna.lab
```

- Generate SSH certificate

```
Switch3(config)#crypto key generate rsa general-keys modulus
```

- Configure a local user account


```
Switch3(config)#username labuser privilege 15 secret labpassw
```

- Enable SSH services

```
Switch3(config)#line vty 0 15
Switch3(config-line)#transport input ssh
Switch3(config-line)#login local
Switch3(config-line)#end
```

- Validate SSH

```
show ip ssh
```

- Test SSH

```
Switch4#ssh -l labuser 192.168.99.3
```

- (Optional) Modify the default connection timeout and authentication retries values

```
Switch3(config)#ip ssh time-out 60
Switch3(config)#ip ssh authentication-retries 4
```

18. Tips 1

```
no ip domain lookup
```

19. Switchport Security

Switch(config-if)#switchport port-security	Enable port security on the interface. Sets a default maximum of one MAC address on this port and action taken upon breach to shutdown
Switch(config-if)#switchport port-	Configure the maximum MAC addresses allowed on this port to N

security maximum N	
Switch(config- if)#switchport port- security violation shutdown	If the maximum MAC addresses is breached, this interface will be immediately disabled and the administrator is informed via SNMP
Switch(config- if)#switchport port- security violation restrict	If the maximum MAC addresses is breached, packets with unknown source addresses will be dropped by the switch and the administrator is informed via SNMP
Switch(config- if)#switchport port- security violation protect	Behaves like restrict mode except no notification is made to the administrator
Switch(config- if)#switchport port- security mac- address sticky	MAC addresses learnt are sticky. Once a MAC address is learnt on a port it is saved and can't be removed from the list of MAC addresses associated with this port. You can disable this by re-issuing this command with "no" at the start of the command.
Switch(config- if)#switchport port- security mac- address xxxx.xxxx.xxxx	Allocate a specific MAC address to the list of MAC addresses of this port. An address can be manually added if the mode is sticky or dynamic

Switch#show mac address-table	Display the current learnt MAC addresses and which ports they are associated with
Switch#show port- security	Display a summary of port security status for the switch
Switch#show port- security interface <interfaceID>	Display the port security status for the nominated interface. Includes the state of port security, the violation mode, and the maximum MAC addresses allowed on this port
Switch#show port- security address	Display which MAC addresses are attached to secure ports on the switch and the status of the secure port
Switch#show interface <interfaceID>	Show detailed configuration and status information for the specified interface

20. Logging Synchronous

```
line console 0
logging synchronous
```

21. Configure an 802.1Q Trunk Between the Switches

- Use DTP to initiate trunking on Gi1/0/5.

```
S3(config)# interface g1/0/5
S3(config-if)# switchport mode dynamic desirable
```

```
S3# show interfaces trunk
```

- Manually configure trunk interface Gi1/0/5.

```
S3(config)# interface g1/0/5
S3(config-if)# switchport mode trunk
```

22. Configure Default Gateways on the switches

- For the switch to deliver packets to devices on VLANs outside the management VLAN, it must send this packet to a default gateway router

```
S3(config)# ip default-gateway a.b.c.d
```

Where a.b.c.d is the address of the gateway router the switch should use

23. Basic Router Configuration

```
show ip route
```

Configure Ethernet Interface settings for Router

```
R1(config)# int g0/0/0
R1(config-if)# description Connection to Switch2 - VLAN 20
R1(config-if)# ip address 192.168.20.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# exit
R1#
```

Ping from router

```
R1# ping 192.168.10.11 source g0/0/0
```

Configure SSH access for Router

```
R1# configure terminal
R1(config)# ip domain-name ccna.lab
R1(config)# username admin privilege 15 secret adminpass
R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
R1(config-line)# exit
R1(config)# crypto key generate rsa general-keys modulus 1024
R1(config)# exit
```

```
S3#ssh -l admin 192.168.10.1
```

Note: Routers only have 5 virtual terminal lines (line vty 0 - 4)

24. Configure The LOOPBACK interface on the router

```
R1(config)# interface lo0
R1(config-if)# description Loopback test interface
R1(config-if)# ip address 209.165.200.225 255.255.255.224
R1(config-if)# end
R1#
```

25. Configure a SUB-INTERFACE for the management VLAN

```
R1(config)# interface g0/0/1.99
R1(config-subif)# description Connection to management VLAN
R1(config-subif)# encapsulation dot1Q 99
R1(config-subif)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# end
```

- The third command configures this sub-interface to be attached to VLAN 99. It is important that the correct number is specified here as this relates to all device configurations

26. Spanning Tree Protocol

```
show spanning-tree
```

- Change port cost

```
S3(config)# interface g1/0/2
S3(config-if)# spanning-tree cost 3
```

- Remove port cost changes

```
S1(config)# interface g1/0/2
S1(config-if)# default spanning-tree cost
```

- Configure primary and secondary root bridge for vlan

```
S1(config)# spanning-tree vlan 1,10,99 root primary
S1(config)#spanning-tree vlan 1-9,11-99 root primary
```

```
S3(config)# spanning-tree vlan 1,10,99 root secondary
```

- Configure STP mode

```
S3(config)# spanning-tree mode rapid-pvst
```

- Configure PortFast and BPDU Guard on access ports.

```
S3(config)# spanning-tree portfast default
```

27. Configure EtherChannel

I. PAgP

```
S3(config)# interface range g1/0/5-6
S3(config-if-range)# channel-group 1 mode desirable
S3(config-if-range)# no shutdown
```

```
S4(config)# interface range g1/0/5-6
S4(config-if-range)# channel-group 1 mode auto
S4(config-if-range)# no shutdown
```

- Configure trunk ports

```
S3(config)# interface port-channel 1
S3(config-if)# switchport mode trunk
S3(config-if)# switchport trunk native vlan 99
```

```
S4(config)# interface port-channel 1
S4(config-if)# switchport mode trunk
S4(config-if)# switchport trunk native vlan 99
```

II. LACP

```
S1(config)# interface range f0/3-4
S1(config-if-range)# switchport mode trunk
S1(config-if-range)# switchport trunk native vlan 99
S1(config-if-range)# channel-group 3 mode active
S1(config-if-range)# no shutdown
```

```
S4(config)# interface range g1/0/3-4
S4(config-if-range)# switchport mode trunk
S4(config-if-range)# switchport trunk native vlan 99
S4(config-if-range)# channel-group 3 mode passive
S4(config-if-range)# no shutdown
```